

 SMART GOVERNANCE & CADASTRAL DIGITAL PUBLIC INFRASTRUCTURE

Project Tract

India’s Federated, Parcel–Centric Land Operating System

When an Indian citizen purchases land, they must physically run between six disconnected government silos —Revenue, Sub–Registrar Deeds, Survey (FMB), Town Planning, Tax, and Courts. Because these records never talk to each other, fraudsters exploit the blind spot, locking **66% of all civil litigation** and freezing **₹16 Lakh Crore in dead capital**.

 **The Sovereign Leap:** Tract unites all six departments around one universal spatial key (ULPIN)—delivering instant conclusive title truth in under **400 milliseconds**.

 [GitHub Repository](#)

 [Watch Video Pitch](#)

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66%

Of All Civil Court Cases in India Are Land Disputes

Inherited from parents to children, dragging on for an average of 20 years across generations.

₹16 Lakh Cr

Trapped in Frozen, Disputed Dead Capital

Unbankable land that cannot be mortgaged, farmed peacefully, or acquired for critical highways.

< 4 Minutes

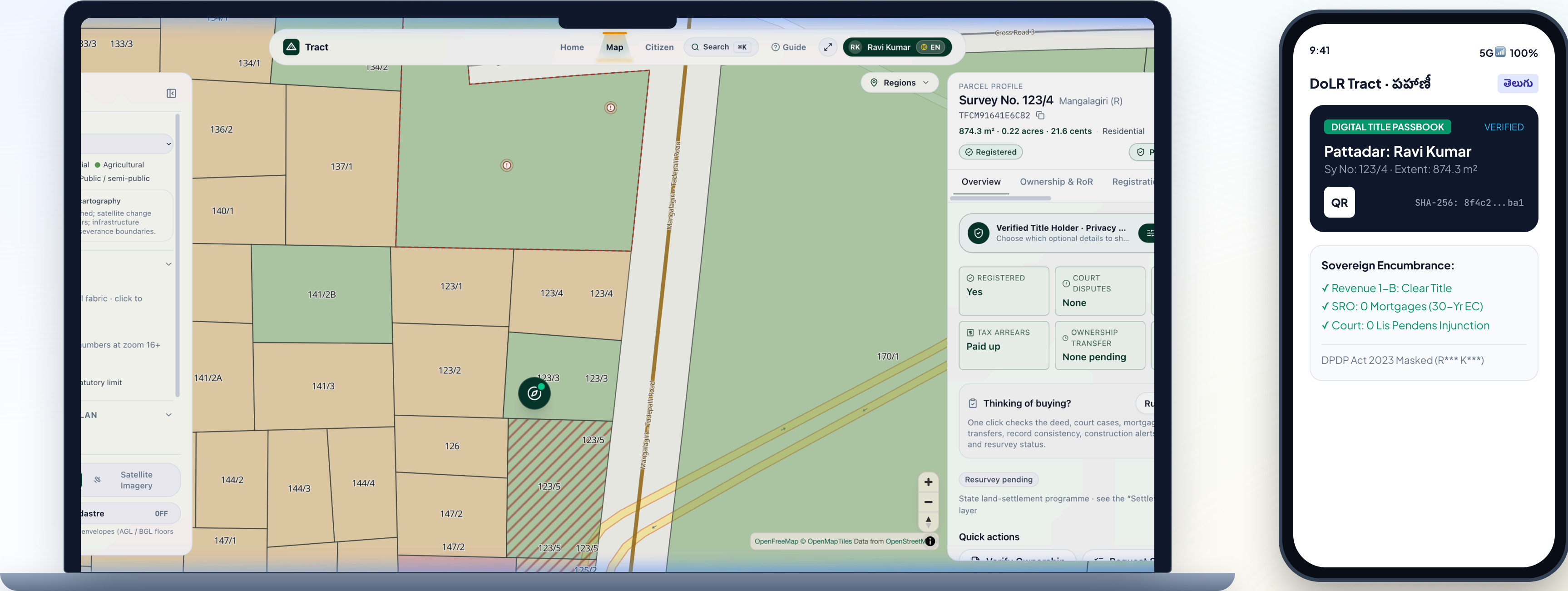
Conclusive Title Verification via ULPIN

Down from 28 days of physical department queues across revenue and registration offices.

FULL SOVEREIGN INTERFACE

The Sovereign Operating System in Action

Live MapLibre GL 3D vector tile shaders paired with offline-first citizen passbooks and field officer consoles.



✓ 6-Department Truth

Synchronizes Revenue, Deeds, FMB & Injunctions in under 400 milliseconds.

◇ ISO 19152 3D Cadastre

Volumetric parcel titling for sky apartments and subterranean metro rail tunnels.

⦿ Severance Math

RFCTLARR 2013 §23A highway corridor severance & 100% solatium compensation.

⬡ Sat Watchdog

Sentinel-2 5-day NDVI vegetation and NDBI construction anomaly pipeline.

End-to-End Architectural Pipeline

Eliminating isolated database silos through an open, provenance-hashed Common Data Model (CLM 1.0 JSON-LD).



Ingests disconnected departmental records across multiple Indian states:

- Revenue:** Meebhoomi (AP), Patta Chitta (TN), Dharani (TG) 1-B Adangals.
- Registration (SRO):** Registered deed instruments & 30-year encumbrance charges.
- Survey:** Village FMB boundary shapefiles & total station traverse points.
- Satellite:** Sentinel-2 Level-2A multispectral 10m imagery.

ADAPTERS: Declarative state dialect mappings translate Telugu, Tamil & Hindi revenue terms.

Enforces mathematical and statutory spatial guardrails directly in the database:

- Cadastral Snap Assistant:** Vertices snap magnetically to authoritative FMB traverse lines.
- ±15% Area Variance Ceiling:** Boundary edits exceeding statutory limits are rejected at the DB level.
- Topological Invariants:** Mathematically eliminates overlapping polygons and slivers.
- Dynamic Vector Tiles:** PostGIS generates binary MVT streams in sub-100ms.

STORAGE: PostgreSQL 16 + PostGIS 3.4 with GiST spatial indexes & geometric partitioning.

Stateless asynchronous core coordinating legal state machines:

- CLM 1.0 JSON-LD:** Universal semantic spatial parcel representation.
- ROR Act §5 State Machine:** Sequential VRO → RI → Tahsildar quasi-judicial conveyor.
- Forensic Deed Scrutiny:** SHA-256 fingerprinting + OCR extent cross-verification.
- DPDP Act 2023:** Purpose-bound consent tokenization & masking.

AI WATCHDOG: 5-day orbital Sentinel-2 pipeline monitors NDVI vegetation drops & NDBI built-up surges.

Empowers citizens, revenue officers, infrastructure, and commercial banks:

- Citizen PWA:** Multilingual search, name change mutation & e-Passbooks.
- Officer Queue:** Automated evidence compilation & digital Speaking Orders.
- PM GatiShakti:** Automated corridor severance math under RFCTLARR 2013 §23A.
- B2B Banking APIs:** Instant 9-point title verification for SBI, HDFC & NABARD.

CLOUD SCALE: Google Cloud Run (asia-south1 Mumbai) + Cloudflare CDN vector tile caching.

\$ PRAGMATIC ECONOMICS & READINESS

Feasibility, Lean Economics & Self-Funding Model

Running a complete Indian state for less than a single district's annual paper and stationery budget.

Monthly Infrastructure Cost (OpEx in Indian Rupees)

DEPLOYMENT TIER	CLOUD COMPONENTS	MONTHLY COST
Pilot Mandal 100,000 Plots	Google Cloud Run (2 vCPU/4GB) + Neon PostGIS + Cloud Storage	~₹8,000 / mo
Full State 30 Million Plots	Autoscaling Cluster + High-Availability PostGIS + Cloudflare Edge CDN	~₹3,80,000 / mo

**A statewide deployment runs for under ₹3.8 Lakhs/month—less than a single district collectorate's annual printing budget!*

3-Tier Self-Funding Monetization Engine

1. **G2G State SaaS Concession:** Annual software license of ₹15 Lakhs – ₹25 Lakhs per district for revenue consoles, satellite alerts, and spatial workflow engines.
2. **G2C Citizen Micro-Fees:** Nominal fee of ₹25 – ₹50 for citizen downloads of digitally signed Title Certificates and FMB maps.
3. **B2B Banking APIs:** High-margin fee of ₹150 – ₹300 per hit charged to commercial banks (SBI, HDFC, NABARD) for automated 9-point title verification.

Operational Readiness & Zero Administrative Friction

- **Direct Statutory Alignment**

Directly enforces Section 5 of the Record of Rights (ROR) Act and State Survey & Boundaries Acts without requiring constitutional amendments.
- **Empowers Existing Revenue Officers**

Does not replace officers. Pre-assembles evidence dossiers for VROs, Surveyors, and Tahsildars to issue valid Speaking Orders.
- **Bandwidth-Efficient for Rural Bharat**

Mapbox Vector Tiles stream over low-bandwidth 3G/4G connections consuming <150 KB per viewport.

Critical Challenges & Statutory Mitigations

- Discrepancy between Revenue and Deeds:** Tract never artificially overwrites records; it surfaces an automated 'Title-Registration Asymmetry' warning to prevent fraudulent sales.
- Boundary Tampering:** Database schema enforces an immutable ±15% statutory area variance ceiling. Any higher variance legally requires gazetted District Collector de-novo approval.
- Privacy vs Public Transparency:** DPDP Act 2023 purpose-bound consent masking protects personal data by default, unmasking strictly with authenticated citizen OTP or bank tokens.

✓ TRANSFORMATIONAL IMPACT

National Economic Multipliers & Human Dignity

Tangible transformation for smallholder farmers, state revenue administrations, infrastructure, and banking.

66% → <10%

Civil Court Land Cases

Disputes eliminated at source by unifying title and registry records.

28d → 4min

Title Verification Time

Instant multi-department search replaces physical queues.

₹16 Lakh Cr

Dead Capital Unlocked

Converts frozen land into bankable collateral for formal credit.

24m → 60d

GatiShakti Land Clearance

Automated RFCTLARR severance math prevents acquisition delays.

 Smallholder Farmers & Rural Families

→ **Escape from 36% Moneylenders:** A verified digital title allows a farmer to secure formal 7% priority-sector bank credit instead of falling into compound debt.

→ **Ending Generational Trauma:** Sparing children from 20-year civil court litigation inherited from parents.

→ **Women Landowner Security:** DPDP Act 2023 masking shields female property owners from harassment and forged transfers.

 State Revenue & Registration Departments

→ **80% Reduction in Mutation Backlogs:** 4-stage desk conveyor pre-assembles statutory evidence for Tahsildars.

→ **Zero Double-Registration Scams:** Sub-Registrar cannot register deed papers without spatial parcel lock.

→ **Automated Speaking Orders:** Standardizes quasi-judicial records to withstand High Court scrutiny.

✓ SOVEREIGN COMPLIANCE

Statutory Grounding & Geospatial Compliance

Built strictly on established parliamentary legislation, OGC open geospatial standards, and national digital infrastructure missions.

1. Statutory Acts

State ROR Acts (§5):
Quasi-judicial mutation conveyor & mandatory Speaking Orders.

RFCTLARR Act 2013 (§23A, §64):
Statutory severance compensation & 100% solatium.

Registration Act 1908:
Provenance-based title vs instrument matrix.

DPDP Act 2023:
Purpose-bound consent masking for citizen data.

2. Geospatial Norms

ISO 19152 (LADM):
Land Administration Domain Model for 3D Volumetric Cadastres.

OGC API Features:
Open geospatial consortium interoperable endpoints.

Mapbox Vector Tiles (MVT):
Binary vector streaming specification 2.1.

W3C JSON-LD 1.1:
Common Linked Data contextual schemas.

3. National Missions

PM GatiShakti:
Multimodal National Master Plan spatial corridor integration.

DILRMP:
Digital India Land Records Modernization Programme.

SVAMITVA Scheme:
Survey of Villages and Mapping with Improved Technology.

MeitY Cloud Guidelines:
Sovereign in-country data residency.

4. Remote Sensing

ESA Copernicus Sentinel-2:
Level-2A surface reflectance satellite pipeline.

NDVI Algorithms:
Rouse et al., Normalized Difference Vegetation Index for encroachment.

NDBI Mapping:
Zha et al., Built-up index for foundation trench detection.

PostGIS 3.4 Specs:
Topological spatial invariant validation algorithms.